

UVLO Lock-Out Latch for an LM5176 Buck-Boost

USB-C PD → 12 V fan-array controller, 42 W — design summary. Full derivations: [uvlo_tlv431.pdf](#) (divider), [uvlo_latch.pdf](#) (latch). Schematic overleaf.

Problem

The control loop is deliberately slow — its crossover sits far below the ~ 160 Hz commutation pulses of the fan load, so the loop does not chase them and the output bank absorbs them instead. With that loop, the LM5176's pre-biased startup does not survive a brief input dropout: on recovery the part restarts into a still-charged output and misbehaves. TI support could not root-cause it and proposed raising the crossover frequency — precluded by the very requirement that set the low crossover in the first place.

Approach

A discrete lock-out latch (four BJTs + a TLV431 shunt reference, all jellybean parts) supplied from V_{OUT} rather than V_{IN} . The TLV431 watches the EN/UVLO divider; when the *input* sags the reference stops conducting, an actively driven “dearm” transistor releases, and a two-transistor SCR core fires and clamps EN/UVLO low. Because the latch is powered from the *output*, it holds until V_{OUT} has decayed to ≈ 1.1 V — so every restart begins from a dead output and the pre-bias path is never exercised. Cold start is safe by construction: at $V_{OUT} = 0$ the latch is unpowered.

Result

(worst case over $\pm 3\%$ resistors, full datasheet spreads, 0–50 °C)

The two rails carry independent threshold sets — input thresholds decide *when the converter runs and when the latch trips*; output thresholds decide *when the latch is able to act at all*, since V_{OUT} is its supply.

V_{IN} domain	divider $R_2/R_{1a}/R_{1b} = 420/150/15\text{ k}\Omega$	turn-on guaranteed in 2.75–4.49 V ; latch trips at $V_{IN} \approx 3.5$ V, inside the IC's own hysteresis window
V_{OUT} domain	latch arms above 2.8 V nom. (2.0–3.6 V worst case)	dearm circuit active from 2.05 V ; EN released once V_{OUT} falls to 1.0–1.8 V
Cost	6.9 mA from the 12 V rail	every resistor a stocked JLCPCB basic 0603 part

The divider was chosen to push the *earliest* possible turn-on as high as it will go (2.75 V) while the *latest* still clears a sagging 4.5 V USB rail — a two-sided constraint on the same ratio, not a single target.

Engineering notes — where the work actually went

- **Correlated vs. independent worst case.** Comparing the *population bands* of the three V_{IN} threshold events makes the design look infeasible. But the latch trip and the IC's own dropout are events of the same device at the same instant: I_{STBY} , I_{HYS} , V_{IN} and R_2 cancel in the difference, leaving a gap set only by V_{TLV} , the divider ratio and V_{th} . Differencing per unit rather than per band is what makes a solution exist at all.
- **A device model that is traceable, not assumed.** No V_{BE} spec exists at the tens-of- μA the latch runs at, so the datasheet $V_{BE(\text{sat})}$ box (0.65–0.85 V at 10 mA) is scaled by $V_T \ln(I_1/I_2)$ down to 0.50–0.70 V at 30 μA . Every threshold is then guaranteed by spec limits alone — no unit screening, no reliance on typicals.
- **Design against a physics floor (an output-rail limit).** The dearm transistor draws its base current from V_{OUT} through the TLV431, so it cannot turn on until $V_{OUT} \geq V_{KA}^{\text{max}} + V_{EB} = 2.02$ V — a hard limit, independent of resistor choice. The design sits at 2.05 V, 27 mV above it, which is what keeps the latch's arm point low and the unprotected output band as narrow as the parts allow.
- **A vendor constraint found on paper, not in smoke.** While latched, the reference's cathode floats to ≈ 15.4 V. A 6 V-rated TLV431 (TI) would sit outside its absolute maximum during normal operation of this circuit; the pin-compatible onsemi part (16 V) is what the design assumes. The same excursion shifts the reference minimum by 21 mV — enough to invalidate an earlier resistor triple that had passed.
- **A revision scrapped on analysis.** The first latch used sub-threshold leakage as its trigger. That current is unspecified and spans $\sim 2000\times$ over part spread and temperature, so it was replaced by a trigger defined purely by resistor ratios and spec'd V_{BE} .

Verification

Two Python tools: closed-form constraint geometry plus an exhaustive corner sweep over stocked part values, emitting **21 pass/fail checks** (0 fail; 2 flagged as bench-review items) and the constraint-region figures. Every threshold is cross-checked against an LTspice transient — simulated engage **3.47 V vs. 3.51 V predicted (1.3 %)**, hold through

a full input re-application at $V_{OUT} = 12\text{ V}$, release at $V_{OUT} = 1.08\text{ V}$ inside the predicted 1.0–1.8 V band. The checker re-reads the `.raw` file, so the simulation and the hand analysis cannot silently drift apart.

Known limit, and the open question it leaves

The latch can only act once V_{OUT} is above its arm point, so **a dropout with V_{OUT} anywhere below $\approx 3.6\text{ V}$ (worst-case unit) is unprotected** — the band runs down to zero, not just to the release point. Two things follow, and the second is not yet closed:

- Body diodes do *not* bound the exposure: the converter switches synchronously, so an enabled FET conducts either way.
- The right failure model is *energy*, not charge, and it is not benign. A passive dump through a conducting FET would merely equalise the two banks at 3.56 V. But a buck-boost moves energy through an inductor, and the input bank is $90\times$ smaller ($C_{OUT} \approx 18.9\text{ mF}$ vs. $C_{IN} \approx 0.21\text{ mF}$), so a reverse transfer is a *boost* into a small capacitor:

$$V_{IN} \approx V_{OUT} \sqrt{C_{OUT}/C_{IN}} = 3.6 \times 9.5 \approx \mathbf{34\text{ V}},$$

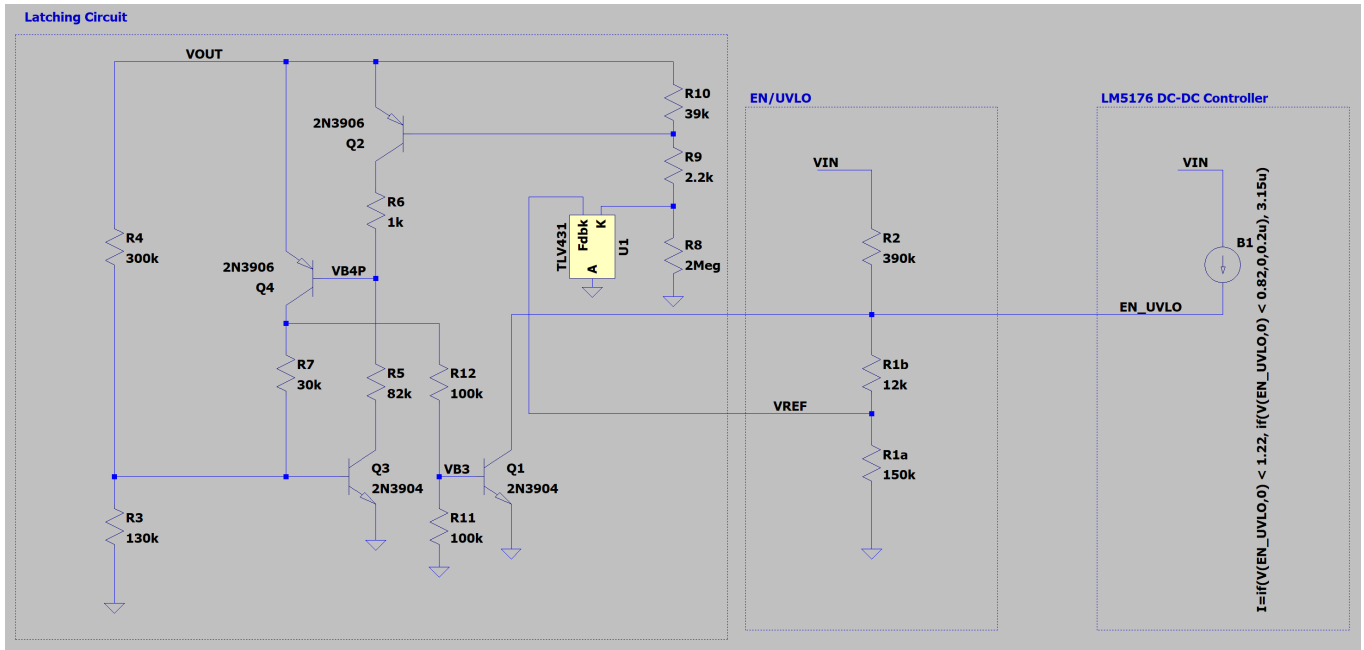
against a 22 V operating maximum and 35 V input electrolytics — and whatever USB source is attached sees it. Even a quarter of the energy reaching the input gives $\approx 17\text{ V}$.

Open item (safety, not housekeeping): confirm by bench and simulation that the LM5176's pre-biased startup performs no reverse energy transfer for V_{OUT} below the arm point. Until that is shown, the low- V_{OUT} band is an accepted, documented exposure — and the case for extending protection below the arm point (or clamping the input) rests on that measurement.

Artifacts: `10a_uvlo_resistor_search.py` (divider search) · `10b_uvlo_latch.py` (latch checks + LTspice cross-check) · two LaTeX design notes · LTspice schematic and transient.

Circuit

Left: the lock-out latch, supplied from V_{OUT} . Right: the LM5176 EN/UVLO divider (R_2 , R_{1b} , R_{1a}) that the TLV431 senses and that Q1 clamps, plus the behavioural source used to model the EN-pin bias current in simulation.



Simulated sequence

Cold start → run → input dropout (latch engages) → input returns while V_{OUT} is still high (latch holds) → V_{OUT} decays (latch releases, clean restart).

UVLO lockout latch v3b -- latest LTspice transient (as-drawn values)

